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Samples of Work

Smart Sustainability Bat Box



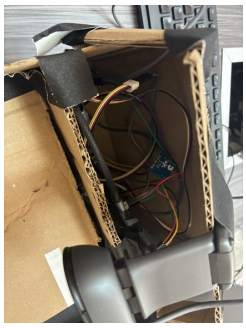
Smart bat box with view of components



Lighting sensor and bottom view



Arduino, battery, and 3 of the sensors



Another angle of the sensors



The outside of the bat box

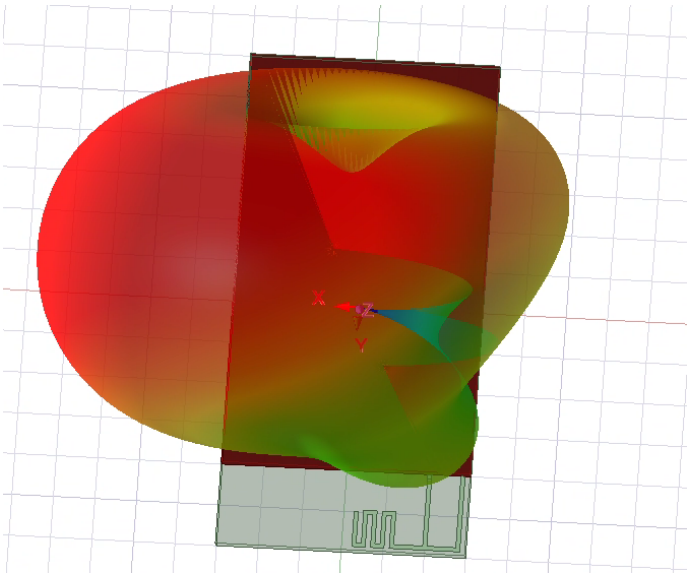
The **Smart Sustainability Bat Box** is a tool to help bat box owners better care for them, and allows for better data collection for citizen science efforts in support of bats and their roosting. The dashboard at batbox.tech provides live temperature, light, sound, and humidity readings and gives owners maintenance and care advice when conditions are outside the safe range for bats. The box also uses computer vision to take photos of and time-stamp bats as they enter and leave, making data collection simple, and provides a live alert if bats are staying on the outside of the box, signaling that something is wrong.

The project was made as an entry into the sustainability track for the MHacks 2026: *Build Something That Grows* hackathon in 24 hours, with a team of four, using hardware provided by the competition and a limited selection of electronics brought by the team. The information currently shown on the dashboard is sample data taken before the end of the hackathon. A demo video is available at youtube.com/watch?v=5mFW1SvJ1cE.

Goal. Citizen science efforts are some of the most important ways people can contribute to benefiting the environment. These efforts inspired us to create technology that can help make them more effective. Tracking bat populations and creating man-made roosts is prevalent in many states such as Michigan and Alaska (bat boxes) in order to support and keep an eye on bats. Bats are facing large amounts of habitat loss and struggling due to climate change. Simultaneously, bats are a necessary and vital part of the ecosystem, as they curb insect populations that can damage farmland and forests. With higher insect populations, farmers tend to use more pesticides and fertilizers, which get swept into our rivers and lakes, causing pollution and harming those ecosystems as well (e.g., Lake Erie's algae blooms harming wildlife through toxin exposure and oxygen deprivation). While declining bat populations aren't the only cause of such problems, making a difference where we can is the best thing we can do to help the environment.

Result. We used an Arduino UNO Q connected to a Logitech webcam, running a bat detection image classification neural network we trained using Edge Impulse. The box takes a photo every time it detects a bat and sends it with a timestamp to our Firebase Realtime Database, which is then displayed by the dashboard. The UNO Q is connected via its pins to the temperature, humidity, light, and sound sensors, whose readings are sent through serial and added to the JSON sent to the database. The bat box itself is put together with cardboard boxes from the Duderstadt Center's loading dock and some duct tape, with the electronics stored at the top. The dashboard not only presents the data from the database, but also gives recommended actions, such as patching gaps or moving the box to a quieter location when the volume or light is too high. If the camera sees bats near the entrance for around 20 minutes, it gives an urgent alert that the user should check on the box, since something is likely keeping the bats from going in.

Bluetooth Antenna



Bluetooth antenna with its radiation pattern

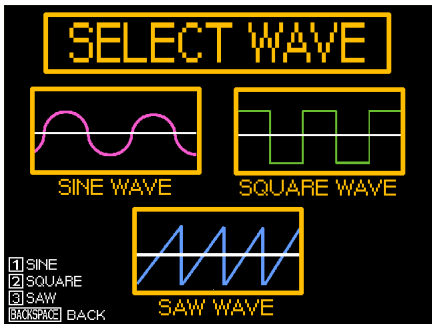
Goal. I was working on our R&D custom radio project for the Michigan Mars Rover team, and I wanted to learn how to use Ansys HFSS to simulate and design the project. I had in mind a plant moisture sensor that would connect to another device via Bluetooth to give live alerts, so the idea was to design and simulate an antenna, optimize it, and use it as the Bluetooth board for the project. While the plant project has been put off (my plant died...), I fully designed the antenna.

Result. I designed an inverted-F antenna in HFSS with a parameter for the length of the smaller horizontal segments, which I then simulated to find the optimal value in the 2.4 GHz range. The result was 90% efficiency over 100 MHz of bandwidth to transfer the data.

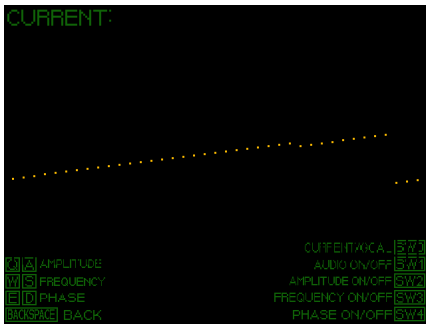
Synth Synthesis



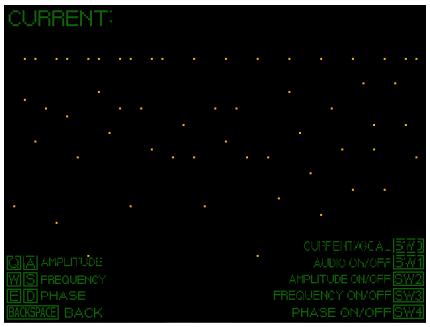
Starting screen and running on the FPGA plus hardware



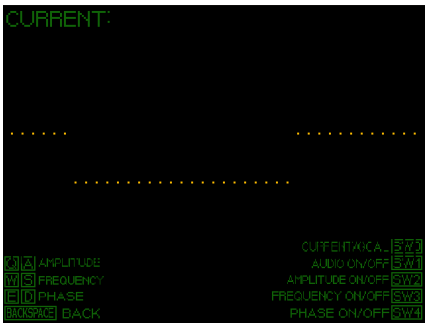
Wave selection screen



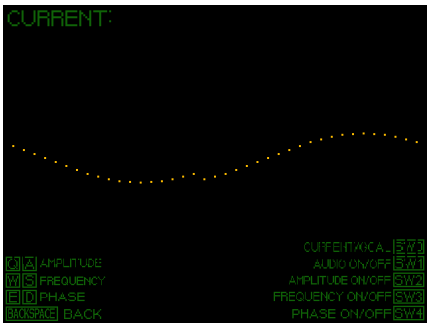
Interactive wave modulation screen with a saw wave



The saw wave with frequency, phase, and amplitude modulated



A square wave



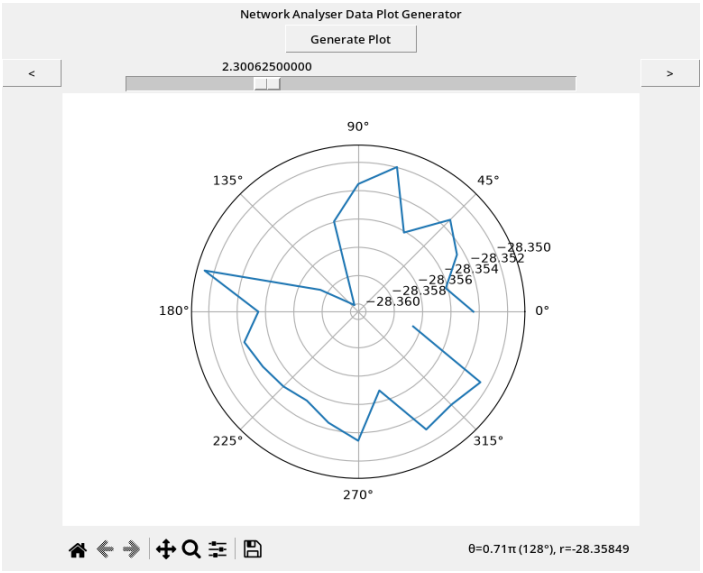
The sine wave

Synth Synthesis is an FPGA-based toy built to introduce children to different wave modulations. It provides children with a variety of goal waveforms and sounds that they then try to match. Children get to play around with and practice making different sounds to build intuition for making their own music.

Goal. It's not necessarily easy to grasp synthesizer modulations when you're between 8 and 10 years old. Many online resources are either too specific to certain tools to build real understanding, or too advanced for beginners. For our class ENGR100.250: Microprocessors and Toys, we were tasked with building an educational toy on an FPGA, and this was our group's submission.

Result. Throughout the class we built up the different pieces that allowed us to create this final project. We used Verilog, a hardware description language, to create our microprocessor with the capability of reading our assembly instruction set from the FPGA memory. We then used the assembly language to create drivers for a display, speaker, and keyboard, which we used to build our project. The project gives the user goal waves and the option to try different modulations and use them together. The user then tries to alter their initial starting wave to match the one presented to them.

360° RF Test Analyzer



Analyzer producing a polar plot at a frequency of 2.300625 GHz

This was a tool I built while shadowing at DENSO International America.

1. It automates the processing of 360° anechoic chamber readings from a vector network analyzer using Python, organizing them by frequency and angle.
2. It then creates a GUI to generate polar plots at each of the 1601+ measured frequencies (a larger range of frequencies can produce even more) to inform decisions.